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United States Patent Application Publication

20250269759

Kind Code

A1

Publication Date

August 28, 2025

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REPLACEABLE BATTERY FIXING STRUCTURE

Abstract

A replaceable battery fixing structure includes a housing, and a lid that opens and closes an opening of the housing, and a fixing mechanism that fixes the lid to the housing. The housing includes a first inclined surface on an upper back portion of an interior thereof against which an upper corner portion of the replaceable battery abuts, and the lid includes a plurality of second inclined surfaces that abut a lower portion of a proximal end of the replaceable battery. The first inclined surface is inclined downward toward the back, and the second inclined surface is inclined forward toward the bottom.

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Family ID: 1000008504656

Appl. No.: 19/062009

Filed: February 25, 2025

Foreign Application Priority Data

JP 2024-028262

Feb. 28, 2024

Publication Classification

Int. Cl.: B60L53/80 (20190101); H01M50/204 (20210101); H01M50/249 (20210101);
H01M50/264 (20210101); H01M50/271 (20210101)

U.S. Cl.:

CPC B60L53/80 (20190201); H01M50/204 (20210101); H01M50/249 (20210101);
H01M50/264 (20210101); H01M50/271 (20210101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-028262 filed on Feb. 28, 2024, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.

TECHNICAL FIELD

[0002] This disclosure relates to the fixing structure of replaceable batteries and the structure of battery-powered vehicles in which replaceable batteries are installed.

BACKGROUND

[0003] JP2022-539061A discloses a battery-powered vehicle that is equipped with replaceable batteries that can be easily attached and removed. JP 2022-539061A also discloses a configuration in which multiple replaceable batteries are housed within the housing attached to the vehicle.

SUMMARY

[0004] When a plurality of replaceable batteries are accommodated in a housing attached to a vehicle, the accommodated replaceable batteries need to be fixed to the housing. In the case of a structure in which the replaceable battery is inserted into the housing in the horizontal direction, a structure in which the replaceable battery is fixed in the horizontal direction by a bolt or the like is often used.

[0005] In this case, the fixing force of the upper and lower parts of the replaceable battery is small, and the fixing of the replaceable battery may be insufficient. In addition, the fixing operation using the bolt takes a long time, and the replacement time of the replaceable battery may become long.

[0006] Accordingly, an object of the present disclosure is to facilitate attachment and detachment of a replaceable battery and to improve a fixing force of the replaceable battery.

[0007] A replaceable battery fixing structure of this disclosure includes a housing that houses a plurality of replaceable batteries in an interior of the housing and includes an opening through which the plurality of replaceable batteries are inserted into the interior of the housing, and a lid that opens and closes the opening, the lid comprising a closing surface that closes the opening, and a fixing mechanism that fixes the lid to the housing. The housing includes a plurality of first fixing surfaces at an upper back portion of the interior of the housing, each of the plurality of first fixing surfaces being configured to abut an upper corner portion of a distal end of each of the plurality of replaceable batteries when each of the plurality of replaceable batteries is inserted into the interior of the housing, the upper back portion of the interior of the housing and the distal end of each of the plurality of replaceable batteries located remote from the opening of the housing. The lid includes a plurality of second fixing surfaces at a lower portion of the closing surface, the plurality of second fixing surfaces being configured to abut a lower portion of a proximal end of each of the plurality of replaceable batteries inserted into the interior of the housing when the opening is closed, the proximal end adjacent to the opening of the housing. Each of the plurality of first fixing surfaces is inclined downward toward a back end of the interior of the housing, the back end of the interior located remote from the opening of the housing. Each of the plurality of second fixing surfaces is inclined forward toward a bottom of the interior of the housing, the forward being toward a front end of the interior of the housing. When the lid is fixed to the housing by the fixing mechanism, each of the first fixing surfaces presses downward the upper corner portion of the distal end of each of the plurality of replaceable batteries in a forward oblique direction, and each of the plurality of second fixing surfaces presses downward the lower portion of the proximal end of each of the plurality of replaceable batteries in a backward oblique direction.

[0008] Thus, the replaceable battery can be simultaneously fixed in the insertion direction into the housing and the vertical direction. In addition, the plurality of replaceable batteries can be fixed to

the housing by a common fixing mechanism. This facilitates the attachment and detachment of the replaceable battery and improves the fixing force of the replaceable battery.

[0009] A replaceable battery fixing structure of this disclosure further may include one or both of a plurality of first elastic members and a plurality of second elastic members. Each of the plurality of first elastic members may be attached to the upper corner portion of the distal end of each of the plurality of replaceable batteries and abuts each of the plurality of first fixing surfaces. Each of the plurality of second elastic members may be attached to the lower portion of the proximal end of each of the plurality of replaceable batteries and abuts each of the plurality of second fixing surfaces.

[0010] Thus, the fixing force of the replaceable battery can be further improved. In addition, it is possible to suppress displacement of the position of the replaceable battery with respect to the housing due to vibration.

[0011] The fixing mechanism of a replaceable battery fixing structure of this disclosure may include a hook provided on a side surface of the housing, and an operation lever rotatably attached to an outer surface of the lid, and a connector connected to the operation lever at a position between a rotation center of the operation lever and an operation end of the operation lever which is at the other end of the operation lever opposite the end with the rotation center. With the opening of the housing being closed by the lid, the lid may be fixed to the housing by rotating the operation lever toward the lid to engage the tip of the connector with the hook such that the hook is pulled to the front side.

[0012] As described above, since the fixing mechanism has a structure that does not include bolt fastening, the attachment/detachment operation of the replaceable battery can be easily performed, and the attachment/detachment time of the replaceable battery to/from the battery-powered vehicle can be shortened.

[0013] A battery electric vehicle of this disclosure includes a vehicle body, and a housing that is attached to the vehicle body, houses a plurality of replaceable batteries in an interior of the housing, and includes an opening through which the plurality of replaceable batteries are inserted into the interior of the housing, and a lid that opens and closes the opening, the lid comprising a closing surface that closes the opening, and a fixing mechanism that fixes the lid to the housing. The housing is attached to the vehicle body such that the opening faces a vehicle side surface, and includes a plurality of first fixing surfaces at an upper back portion of the interior of the housing, each of the plurality of first fixing surfaces being configured to abut an upper corner portion of a distal end of each of the plurality of replaceable batteries when each of the plurality of replaceable batteries is inserted into the interior of the housing along a vehicle width direction from the opening, the upper back portion of the interior of the housing and the distal end of each of the plurality of replaceable batteries located remote from the opening of the housing. The lid includes a plurality of second fixing surfaces at a lower portion of the closing surface, the plurality of second fixing surfaces being configured to abut a lower portion of a proximal end of each of the plurality of replaceable batteries inserted into the interior of the housing when the opening is closed, the proximal end adjacent to the opening of the housing. Each of the first fixing surfaces is inclined downward toward a back end of the interior, the back end of the interior located remote from the opening of the housing. Each of the plurality of second fixing surfaces is inclined forward toward a bottom of the interior, the forward being toward a front end of the interior. When the lid is fixed to the housing by the fixing mechanism, each of the first fixing surfaces presses downward the upper corner portion of the distal end of each of the plurality of replaceable batteries in a forward oblique direction, and each of the plurality of second fixing surfaces presses downward the lower portion of the proximal end of each of the plurality of replaceable batteries in a backward oblique direction.

[0014] Thus, the replaceable battery can be simultaneously fixed in the insertion direction into the housing and the vertical direction. In addition, the plurality of replaceable batteries can be fixed to the housing by a common fixing mechanism. In addition, in the battery-powered vehicle, it is

possible to easily attach and detach the replaceable battery and to improve the fixing force of the replaceable battery. In addition, the replaceable battery can be attached and detached from the side surface of the battery-powered vehicle.

[0015] The lid of a battery electric vehicle of this disclosure may be rotatably attached around a rotation shaft attached to the vehicle body substantially in parallel with a lower side of the opening of the housing. The fixing mechanism of a battery electric vehicle of this disclosure may include a hook provided on an upper portion of a side surface of the housing, and an operation lever rotatably attached to an outer surface of the lid, and a connector connected to the operation lever at a position between a rotation center of the operation lever and an operation end of the operation lever which is at the other end of the operation lever opposite the end with the rotation center.

[0016] With the opening of the housing being closed by the lid, the lid may be fixed to the housing by rotating the operation lever toward the lid to engage the tip of the connector with the hook such that the hook is pulled to the front side.

[0017] As described above, since the fixing mechanism has a structure that does not include bolt fastening, the attachment/detachment operation of the replaceable battery can be easily performed, and the attachment/detachment time of the replaceable battery to/from the battery-powered vehicle can be shortened.

[0018] According to the present disclosure, it is possible to facilitate attachment and detachment of the replaceable battery and to improve the fixing force of the replaceable battery.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0019] FIG. 1 is a right side view of an electric vehicle including a replaceable battery fixing structure according to an embodiment;

[0020] FIG. 2 is an elevational cross-sectional view of the electric vehicle including the replaceable battery fixing structure of the embodiment, and is a cross-sectional view taken along line A-A of FIG. 1;

[0021] FIG. 3 is a cross-sectional view of the housing and the replaceable battery of the replaceable battery fixing structure of the embodiment, and is a detailed view of a portion B shown in FIG. 2;

[0022] FIG. 4 is a cross-sectional view of the housing, the lid, and the replaceable battery of the replaceable battery fixing structure of the embodiment, and is a detailed view of a portion C shown in FIG. 2;

[0023] FIG. 5 is a plan view showing a fixing mechanism of the replaceable battery fixing structure of the embodiment.

DESCRIPTION OF EMBODIMENTS

[0024] Hereinafter, a replaceable battery fixing structure **100** according to an embodiment will be described with reference to the drawings. Note that FR, UP, and RH shown in the drawings indicate a front side, an upper side, and a right side of the electric vehicle **200** including the replaceable battery fixing structure **100**, respectively. The opposite directions of FR, UP, and RH indicate the rear side, the lower side, and the left side, respectively. Hereinafter, in the case of simply using the front-rear direction, the left-right direction, and the up-down direction, unless otherwise specified, the front-rear direction, the left-right direction, and the up-down direction of the electric vehicle **200** are indicated. The electric vehicle **200** is a battery electric vehicle that runs by driving an electric unit (not shown) with electric power supplied from the replaceable battery **70** charged by an external power source.

[0025] First, a configuration of a vehicle body **30** of an electric vehicle **200** including a replaceable battery fixing structure **100** will be described with reference to FIG. 1. The vehicle body **30** includes a frame **10**, a front compartment **31**, a cab **32**, and a deck **33**. The electric vehicle **200** is a

frame-type electric truck.

[0026] As shown in FIGS. **1** and **2**, the frame **10** is a ladder-type frame member including a side rail **11**, first to fifth cross members **12**, **13**, **14**, **15**, and **16**, and first and second battery supports **17** and **18**. The first to fifth cross members **12**, **13**, **14**, **15**, and **16** are longitudinal members extending in the left-right direction, and connect the left and right side rail **11**. The first and second battery supports **17** and **18** are longitudinal members that connect the left and right side rail **11** between the cab **32** and the third cross member **14**.

[0027] A front suspension bracket **21** is attached to the vehicle width direction outer side of the side rail **11** between the first cross member **12** and the second cross member **13** on the vehicle front side. A suspension device (not shown) for the front wheel **35** is attached to the front suspension bracket **21**. A front compartment **31** is disposed in an upper portion between the first cross member **12** and the second cross member **13**. A suspension device (not shown) for the rear wheel **36** is attached between the third cross member **14** and the fourth cross member **15** on the rear side of the vehicle.

[0028] A front cab mount bracket **22** is attached to the vehicle width direction outer side of the side rail **11** behind the second cross member **13**. A front cab mount **27** that supports the front of the cab **32** is mounted on the front cab mount bracket **22**. A rear cab mount bracket **23** is attached to the side rail **11** behind the front cab mount **27**. A rear cab mount **28** that supports the rear of the cab **32** is mounted on the rear cab mount bracket **23**. A first deck mount **24**, a second deck mount **25**, and a third deck mount **26** are attached to an upper portion of the side rail **11** behind the rear cab mount bracket **23**. The first through third deck mounts **24**, **25**, **26** support the deck **33**.

[0029] Next, the replaceable battery fixing structure **100** will be described with reference to FIGS. **1** to **4**. As shown in FIG. **1**, the replaceable battery fixing structure **100** includes a housing **40**, a lid **50**, and a fixing mechanism **60**.

[0030] As shown in FIGS. **1** and **2**, the housing **40** houses, in the interior **42A**, a replaceable battery **70** that is charged by an external power source and supplies power to an electric drive unit (not shown). The housing **40** includes a housing main body **41**, an opening **42B**, a front flange **43**, a back plate **45**, a receiving connector **46**, and a back bracket **47**.

[0031] The housing main body **41** is a rectangular longitudinal member having a rectangular closed cross section. The housing main body **41** is attached to the vehicle body **30** so that the opening **42B** faces the right side surface of the electric vehicle **200**. The right end of the housing main body **41** is a rectangular opening **42B**. The left end of the housing main body **41** is closed by a back plate **45**. The housing main body **41** houses three replaceable batteries **70** in the interior **42A**. The replaceable battery **70** is inserted into the interior **42A** from the opening **42B** along the vehicle width direction. In the following description, the side on which the opening **42B** is disposed is referred to as the front side of the housing **40**, and the side away from the opening **42B** is referred to as the back side of the housing **40**. The insertion end of the replaceable battery **70** into the interior **42A** is referred to as a distal end **70A**, and the opposite end is referred to as a proximal end **70B**. The proximal end **70B** is the withdrawal end when the replaceable battery **70** is withdrawn from the interior **42A**.

[0032] A front flange **43** is attached to a peripheral edge of the opening **42B** of the housing main body **41**. The front flange **43** includes an upper flange **43A** and a lower flange **43B**. A packing **44** is set outside the upper flange **43A** and outside the lower flange **43B**. The back plate **45** is a plate member that closes the back side of the housing main body **41**. A receiving connector **46** to which a connector **78** of the replaceable battery **70** is connected is attached to the center of the back plate **45**. The receiving connector **46** is connected to an electric unit (not shown). Further, a back bracket **47** is attached to an upper rear portion of the interior **42A** of the housing main body **41**. Details of the back bracket **47** will be described later with reference to FIG. **3**.

[0033] The replaceable battery **70** is a longitudinal member having a quadrangular cross section. The replaceable battery **70** includes a battery body **71** that is a combination of a plurality of battery

cells, a casing **72** that houses the battery body **71**, a connector **78**, and a front portion **73**. A connector **78** is provided at the distal end **70A** of the replaceable battery **70**. A first elastic member **76** is attached to the upper corner portion **72A** of the distal end **70A** of the replaceable battery **70**. The first elastic member **76** is made of, for example, rubber, resin, or the like. Details of the upper corner portion **72A** and the first elastic member **76** will be described later together with a detailed structure of the back bracket **47** with reference to FIG. 3.

[0034] A front portion **73** is attached to the proximal end **70B** of the replaceable battery **70**. The front portion **73** includes a handle **74** and a lower portion **75**. The handle **74** is disposed in an upper portion of the front portion **73**. The handle **74** is a portion that is gripped when the replaceable battery **70** is pulled out from the housing **40**. The lower portion **75** is disposed at a lower portion of the front portion **73**. The lower portion **75** is a portion located near the lower portion of the opening **42B** when the replaceable battery **70** is inserted into the interior **42A** of the housing **40**. A second elastic member **77** is attached to the front side of the lower portion **75**. The second elastic member **77** is made of, for example, rubber, resin, or the like. Details of the lower portion **75** and the second elastic member **77** will be described later with reference to FIG. 4.

[0035] The lid **50** is rotatably attached to the vehicle body **30** by a hinge **55** attached to the vehicle body **30** substantially in parallel with the lower side of the opening **42B** of the housing **40**. As indicated by an arrow **91** in FIG. 2 and an arrow **92** in FIG. 4, the lid **50** rotates around the rotation shaft **56** of the hinge **55** to open and close the opening **42B**. A lid **50** indicated by a broken line in FIG. 2 shows a state in which the lid **50** is opened, and a lid **50** indicated by a solid line shows a state in which the lid **50** closes the opening **42B**.

[0036] The lid **50** includes an outer plate **51** and an inner plate **52**. The peripheral edge portions of the inner plate **52** and the outer plate **51** overlap each other to form a lid flange **54**. The lid flange **54** includes a lid upper flange **54A** and a lid lower flange **54B**. The packing **44** is sandwiched between the lid flange **54** and the front flange **43** of the housing **40** when the lid **50** closes the opening **42B**. An annular bead **51B** is provided on the inner peripheral side of the lid flange **54** of the outer plate **51**. The bead **51B** has a groove-shaped cross-sectional shape protruding outward.

[0037] As shown in FIG. 4, the hinge **55** includes a base **57** fixed on the bracket **19** mounted on the right side rail **11**, a rotation shaft **56** mounted on the base **57**, and a hinge plate **58** rotatably mounted around the rotation shaft **56**. The hinge plate **58** is fixed to the outer surface of the bead **51B** on the lower side of the lid **50**. Thus, the lid **50** is rotatably attached around the rotation shaft **56** of the hinge **55** attached to the vehicle body **30** to open and close the opening **42B**.

[0038] The inner surface **52A** of the inner plate **52** of the lid **50** constitutes a closing surface that faces the opening **42B** and closes the opening **42B** when the lid **50** closes the opening **42B**. A plurality of second inclined surfaces **53**, which are set at the lower corner portions **75** of the plurality of replaceable batteries **70** and come into contact with the second elastic members **77** when the lid **50** closes the opening **42B**, are provided at a lower portion of the inner surface **52A**. Details of the second inclined surface **53** will be described later together with detailed structures of the lower portion **75** and the second elastic member **77** with reference to FIG. 4.

[0039] A fixing mechanism **60** for fixing the lid **50** to the housing **40** is provided on an upper portion of the outer plate **51** of the lid **50**. Details of the fixing mechanism **60** will be described later with reference to FIG. 5.

[0040] Next, the back bracket **47** of the housing **40**, the upper corner portion **72A** of the replaceable battery **70**, and the first elastic member **76** will be described in detail with reference to FIG. 3. As shown in FIG. 3, the back bracket **47**, the upper corner portion **72A**, and the first elastic member **76** are located away from the opening **42B**.

[0041] As shown in FIG. 3, the back bracket **47** is attached to the rear portion of the upper plate **41A** of the housing main body **41**. The back bracket **47** is a mountain-shaped folded plate member, and includes a first inclined surface **48** inclined downward toward the rear of the housing **40**.

[0042] The upper corner portion **72A** of the replaceable battery **70** is the upper corner of the distal

end 70A of the casing 72. The first elastic member 76 is attached to the upper corner portion 72A. The upper surface 76A of the first elastic member 76 is inclined downward toward the distal end 70A of the replaceable battery 70. The upper surface 76A of the first elastic member 76 is parallel to the first inclined surface 48 of the back bracket 47. As indicated by white arrows in FIG. 3, when the replaceable battery 70 is inserted into the interior 42A of the housing 40, the upper surface 76A of the first elastic member 76 abuts against the first inclined surface 48 of the back bracket 47. The first inclined surface 48 fixes the upper corner portion 72A of the replaceable battery 70 to the housing main body 41. Accordingly, the first inclined surface 48 is a first fixing surface that fixes the upper corner portion 72A of the replaceable battery 70 to the housing main body 41 when the replaceable battery 70 is inserted into the interior 42A of the housing 40.

[0043] When three replaceable batteries 70 are accommodated in the housing main body 41 as in the replaceable battery fixing structure 100 of the present embodiment, a total of three back brackets 47 are provided for each of the replaceable batteries 70.

[0044] Next, a detailed structure of the second inclined surface 53 of the lid 50, the lower portion 75 of the replaceable battery 70, and the second elastic member 77 will be described with reference to FIG. 4.

[0045] Next, a detailed structure of the second inclined surface 53 of the lid 50, the lower portion 75 of the replaceable battery 70, and the second elastic member 77 will be described with reference to FIG. 4.

[0046] When three replaceable batteries 70 are accommodated in the housing main body 41 as in the replaceable battery fixing structure 100 of the present embodiment, a total of three second inclined surfaces 53 are provided for each of the replaceable batteries 70 in the lid 50.

[0047] Next, the fixing mechanism 60 will be described with reference to FIG. 5. The fixing mechanism 60 includes a base 61, a hook 64, an operation lever 63, and a connector 65. As shown in FIG. 5, the hook 64 is attached to the side surface 41C of the housing main body 41. The hook 64 is a folded plate member in which a U-shaped engagement claw 64A is formed at a tip end thereof. The base 61 is attached to the outer surface 51A of the outer plate 51 of the lid 50, and the base 61 includes a fixing portion 61A fixed to the outer surface 51A of the outer plate 51 and a guide groove 61B for guiding the connector 65. The operation lever 63 is rotatably attached around the rotation shaft 62 attached to the fixing portion 61A. Here, the center of the rotation shaft 62 constitutes the rotation center of the operation lever 63. The connector 65 is rotatably attached to the hole 63B closer to the operating end 63A than the rotation shaft 62.

[0048] Next, closing of the lid 50 and fixing of the replaceable battery 70 will be described. When the lid 50 is closed, the engagement claw 64A of the hook 64 protrudes to the outside of the lid 50 through the through hole 59 provided in the lid 50. Then, the distal end of the connector 65 is engaged with the engagement claw 64A of the hook 64, and the operating end 63A of the operation lever 63 is rotated toward the lid 50 as indicated by an arrow 93 in FIG. 5. Then, the connector 65 moves while being guided by the guide groove 61B, and pulls up the engagement claw 64A of the hook 64 to the front side as indicated by an arrow 94 in FIG. 5. As a result, the packing 44 between the lid flange 54 of the lid 50 and the front flange 43 of the housing 40 collapses, and the lid 50 closes the opening 42B. Then, when the operation lever 63 is rotated until it becomes parallel to the outer plate 51 of the lid 50, the state in which the connector 65 pulls up the hook 64 is held. As described above, when the lid 50 is closed, the lid 50 presses the replaceable battery 70 toward the back of the housing 40 as indicated by the white arrows in FIGS. 3, 4, and 5. When three replaceable batteries 70 are accommodated in the housing main body 41 as in the replaceable battery fixing structure 100 of the present embodiment, when the lid 50 is closed, the lid 50 simultaneously presses the three replaceable batteries 70 toward the back of the housing 40.

[0049] As indicated by white arrows in FIG. 3, when the replaceable battery 70 is pressed toward the back of the housing 40, the first elastic member 76 attached to the upper corner portion 72A of the replaceable battery 70 abuts against the first inclined surface 48 of the back bracket 47. Then,

the first fixing force F in the vertical direction with respect to the first inclined surface **48** is applied to the upper corner portion **72A** via the first elastic member **76**. The first fixing force F presses the upper corner portion **72A** of the replaceable battery **70** obliquely downward toward the front. Accordingly, a horizontal force F1 for pressing the replaceable battery **70** toward the front side and a downward force F2 for pressing the replaceable battery **70** downward are applied to the replaceable battery **70**.

[0050] As shown in FIG. **4**, when the lid **50** closes the opening **42B**, the second inclined surface **53** of the lid **50** contacts the second elastic member **77**. Then, the second fixing force G in the vertical direction with respect to the second inclined surface **53** is applied to the lower portion **75** of the replaceable battery **70** via the second elastic member **77**. The second fixing force G presses the lower portion **75** of the replaceable battery **70** downward in the diagonally rearward direction. Accordingly, a horizontal force G1 for pressing the replaceable battery **70** toward the back and a downward force G2 for pressing the replaceable battery **70** downward are applied to the replaceable battery **70**.

[0051] When the lid **50** closes the opening **42B**, the replaceable battery **70** is fixed between the back bracket **47** and the lid **50** by the horizontal forces F1 and G1. The replaceable battery **70** is fixed between the bottom plate **41B** of the housing main body **41** and the first inclined surface **48** and the second inclined surface **53** by the downward forces F2 and G2.

[0052] As described above, the replaceable battery fixing structure **100** can simultaneously fix the replaceable battery **70** in the insertion direction into the housing **40** and in the vertical direction. Thus, the fixing force of the replaceable battery **70** can be improved. Further, by fixing the common lid **50** to the housing **40** by the fixing mechanism **60**, it is possible to fix the three replaceable batteries **70** to the housing **40** at one time, and it is possible to shorten the time for attaching and detaching the replaceable batteries **70** to and from the electric vehicle **200**.

[0053] In addition, since the fixing mechanism **60** includes the base **61**, the hook **64**, the operation lever **63**, and the connector **65** and does not include bolting, it is possible to easily attach and detach the replaceable battery **70**. In addition, it is possible to shorten the time for attaching and detaching the replaceable battery **70** to and from the electric vehicle **200**.

[0054] Further, in the replaceable battery fixing structure **100**, since the first elastic member **76** and the second elastic member **77** are provided, the fixing force of the replaceable battery **70** can be further improved. In addition, it is possible to suppress displacement of the position of the replaceable battery **70** with respect to the housing **40** due to vibration.

[0055] In addition, in the electric vehicle **200**, since the housing **40** is attached to the vehicle body **30** so that the opening **42B** faces the vehicle side surface, the replaceable battery **70** can be attached and detached from the side surface of the electric vehicle **200**.

[0056] In the above description, the replaceable battery **70** includes the first elastic member **76** in the upper corner portion **72A** and the second elastic member **77** in the lower portion **75**, but may include only one of the first elastic member **76** and the second elastic member **77**. When the upper corner portion **72A** does not include the first elastic member **76**, the first inclined surface **48** of the back bracket **47** abuts on the upper corner portion **72A**. Similarly to the upper surface **76A** of the first elastic member **76**, the upper corner portion **72A** may be configured to be parallel to the first inclined surface **48**. Similarly, when the lower portion **75** does not include the second elastic member **77**, the front surface of the lower portion **75** may be configured to be parallel to the second inclined surface **53** similarly to the front surface **77A** of the second elastic member **77**.

[0057] In the above description, the lid **50** is rotatably attached to the vehicle body **30** by the hinge **55**. For example, the lid **50** may be configured to be detachable from the housing **40** without being attached to the vehicle body **30**. In this case, the fixing mechanism **60** may be disposed on the upper portion and the lower portion of the lid **50**.

[0058] In the replaceable battery fixing structure **100**, the housing main body **41** is attached to the vehicle body **30** such that the opening **42B** is on the right side surface of the electric vehicle **200**,

but the present disclosure is not limited thereto. For example, the housing main body **41** may be mounted on the vehicle body **30** such that the opening **42B** is on the left side surface of the electric vehicle **200**. The housing main body **41** may be attached to the vehicle body **30** such that the opening **42B** is located behind the electric vehicle **200**. In this case, the replaceable battery **70** is inserted in the front-rear direction of the electric vehicle **200**.

[0059] In the above description, the electric vehicle **200** is described as a frame-type electric truck including the frame **10**, the front compartment **31**, the cab **32**, and the deck **33**. The electric vehicle **200** may be a frame-type SUV or an electric vehicle having a monocoque body. The electric vehicle **200** may be a bus or the like, or may be a light van.

Claims

1. A replaceable battery fixing structure comprising: a housing that houses a plurality of replaceable batteries in an interior of the housing and includes an opening through which the plurality of replaceable batteries are inserted into the interior of the housing; a lid that opens and closes the opening, the lid comprising a closing surface that closes the opening; and a fixing mechanism that fixes the lid to the housing, wherein the housing includes a plurality of first fixing surfaces at an upper back portion of the interior of the housing, each of the plurality of first fixing surfaces being configured to abut an upper corner portion of a distal end of each of the plurality of replaceable batteries when each of the plurality of replaceable batteries is inserted into the interior of the housing, the upper back portion of the interior of the housing and the distal end of each of the plurality of replaceable batteries located remote from the opening of the housing, the lid includes a plurality of second fixing surfaces at a lower portion of the closing surface, the plurality of second fixing surfaces being configured to abut a lower portion of a proximal end of each of the plurality of replaceable batteries inserted into the interior of the housing when the opening is closed, the proximal end adjacent to the opening of the housing, each of the plurality of first fixing surfaces is inclined downward toward a back end of the interior of the housing, the back end of the interior located remote from the opening of the housing, each of the plurality of second fixing surfaces is inclined forward toward a bottom of the interior of the housing, the forward being toward a front end of the interior of the housing, when the lid is fixed to the housing by the fixing mechanism, each of the first fixing surfaces presses downward the upper corner portion of the distal end of each of the plurality of replaceable batteries in a forward oblique direction, and each of the plurality of second fixing surfaces presses downward the lower portion of the proximal end of each of the plurality of replaceable batteries in a backward oblique direction.

2. The replaceable battery fixing structure according to claim 1, further comprising: one or both of a plurality of first elastic members and a plurality of second elastic members, wherein each of the plurality of first elastic members is attached to the upper corner portion of the distal end of each of the plurality of replaceable batteries and abuts each of the plurality of first fixing surfaces, and each of the plurality of second elastic members is attached to the lower portion of the proximal end of each of the plurality of replaceable batteries and abuts each of the plurality of second fixing surfaces.

3. The replaceable battery fixing structure according to claim 1, wherein the fixing mechanism includes, a hook provided on a side surface of the housing, an operation lever rotatably attached to an outer surface of the lid, and a connector connected to the operation lever at a position between a rotation center of the operation lever and an operation end of the operation lever which is at the other end of the operation lever opposite the end with the rotation center, with the opening of the housing being closed by the lid, the lid is fixed to the housing by rotating the operation lever toward the lid to engage the tip of the connector with the hook such that the hook is pulled to the front side.

4. The replaceable battery fixing structure according to claim 2, wherein the fixing mechanism

includes, a hook provided on a side surface of the housing, an operation lever rotatably attached to an outer surface of the lid, and a connector connected to the operation lever at a position between a rotation center of the operation lever and an operation end of the operation lever which is at the other end of the operation lever opposite the end with the rotation center, with the opening of the housing being closed by the lid, the lid is fixed to the housing by rotating the operation lever toward the lid to engage the tip of the connector with the hook such that the hook is pulled to the front side.

5. A battery electric vehicle comprising: a vehicle body; a housing that is attached to the vehicle body, houses a plurality of replaceable batteries in an interior of the housing, and includes an opening through which the plurality of replaceable batteries are inserted into the interior of the housing; a lid that opens and closes the opening, the lid comprising a closing surface that closes the opening; and a fixing mechanism that fixes the lid to the housing, wherein the housing is attached to the vehicle body such that the opening faces a vehicle side surface, and includes a plurality of first fixing surfaces at an upper back portion of the interior of the housing, each of the plurality of first fixing surfaces being configured to abut an upper corner portion of a distal end of each of the plurality of replaceable batteries when each of the plurality of replaceable batteries is inserted into the interior of the housing along a vehicle width direction from the opening, the upper back portion of the interior of the housing and the distal end of each of the plurality of replaceable batteries located remote from the opening of the housing, the lid includes a plurality of second fixing surfaces at a lower portion of the closing surface, the plurality of second fixing surfaces being configured to abut a lower portion of a proximal end of each of the plurality of replaceable batteries inserted into the interior of the housing when the opening is closed, the proximal end adjacent to the opening of the housing, each of the first fixing surfaces is inclined downward toward a back end of the interior, the back end of the interior located remote from the opening of the housing, each of the plurality of second fixing surfaces is inclined forward toward a bottom of the interior, the forward being toward a front end of the interior, when the lid is fixed to the housing by the fixing mechanism, each of the first fixing surfaces presses downward the upper corner portion of the distal end of each of the plurality of replaceable batteries in a forward oblique direction, and each of the plurality of second fixing surfaces presses downward the lower portion of the proximal end of each of the plurality of replaceable batteries in a backward oblique direction.

6. The battery electric vehicle according to claim 5, wherein the lid is rotatably attached around a rotation shaft attached to the vehicle body substantially in parallel with a lower side of the opening of the housing, the fixing mechanism includes: a hook provided on an upper portion of a side surface of the housing, an operation lever rotatably attached to an outer surface of the lid, and a connector connected to the operation lever at a position between a rotation center of the operation lever and an operation end of the operation lever which is at the other end of the operation lever opposite the end with the rotation center, with the opening of the housing being closed by the lid, the lid is fixed to the housing by rotating the operation lever toward the lid to engage the tip of the connector with the hook such that the hook is pulled to the front side.
